

Postblock Effective-Rank Compression Reopens the Directional Route

Singular-Value State Reduction and Full-Future Observability Transfer

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Abstract

RH-76 showed that the production source does not concentrate on one shrinking phase arc. This paper tests the surviving weighted alternative: after the log-square block horizon, does the actual directional state become low rank?

Let $B = A^M X$ and

$$O_M = \sum_{r=0}^{M-1} (A^r)^* Y^* Y A^r.$$

If $q = \|A^M\|_2 < 1$, the full observability Gramian satisfies

$$\|O\|_2 \leq \frac{\|O_M\|_2}{1 - q^2}.$$

For any rank- r approximation B_r , the complete future Hardy energies obey

$$|T(B) - T(B_r)| \leq \sqrt{\frac{\|O_M\|_2}{1 - q^2}} \|B - B_r\|_F.$$

Together with Eckart–Young, this gives a fully nonnormal effective-rank compression theorem: low singular rank of the postblock state, rather than global phase localization, is enough to reduce the future directional problem.

We embed the RH-70 frozen operator, source, and observation arrays as exact dyadic Arb matrices. At all five scales and in both channels, Arb recomputes $A^M X$, validates binary64 rank-1, 2, 4 SVD candidates, accumulates a rigorous one-block observability upper, and transfers each residual through the full future. Rank two captures at least 99% of postblock energy in all ten channels; rank four captures at least 99.9999%. The largest participation rank is 1.868. The largest rank-four full-future Hardy perturbation is 5.35×10^{-6} .

Thus the negative single-arc result does not block the Stage-A route. The observed mechanism is radial/nonnormal focusing into a tiny postblock singular subspace. The open problem is now an analytic all-dyadic singular-value decay theorem and its transport from frozen to analytic triples. Uniform Stage A1, unconditional Stage A4, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: effective rank; singular values; observability Gramian; Hardy energy; block contraction; interval arithmetic.

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1 Introduction

The RH-75 block law identified a sufficient all-level target, while RH-76 closed one proposed mechanism: the source-weighted phases do not occupy a shrinking global arc [3]. That negative

result concerned phase geometry before the block horizon. It did not inspect the singular geometry of the actual state $A^M X$ after radial damping and nonnormal mixing.

This distinction matters. A source may occupy every phase coordinate and still evolve into a low-rank matrix because its columns become aligned. Such alignment is basis invariant and survives broad phase support. Moreover, it can be inserted directly into the Hardy tail through observability, without a normal approximation.

The contribution is twofold:

1. an exact block-observability theorem transferring any low-rank postblock approximation through the complete future;
2. a five-scale interval validation showing unexpectedly strong rank-two and rank-four concentration in the production states.

2 Postblock state and future Hardy map

Let $A \in \mathbb{C}^{n \times n}$, source $X \in \mathbb{C}^{n \times m}$, and observation $Y \in \mathbb{C}^{p \times n}$. For a block horizon M , set

$$B = A^M X, \text{quad} O_M = \sum_{r=0}^{M-1} (A^r)^* Y^* Y A^r.$$

The future Hardy energy launched from a state Z is

$$T(Z)^2 = \sum_{\ell \geq 0} \|Y A^\ell Z\|_F^2 = \text{tr}(Z^* O Z), \quad (1)$$

where O is the full observability Gramian.

Theorem 2.1 (Block observability upper). *If $q = \|A^M\|_2 < 1$, then*

$$O = \sum_{b \geq 0} (A^{*M})^b O_M (A^M)^b, \quad \|O\|_2 \leq \frac{\|O_M\|_2}{1 - q^2}. \quad (2)$$

Proof. Write every time index uniquely as $\ell = bM + r$ with $0 \leq r < M$ and regroup the positive series. Taking norms and summing the geometric series proves the upper. $\square$

Theorem 2.2 (Full-future low-rank transfer). *For any approximation B_r of rank at most r ,*

$$|T(B) - T(B_r)| \leq \sqrt{\frac{\|O_M\|_2}{1 - q^2}} \|B - B_r\|_F. \quad (3)$$

The smallest possible Frobenius residual is

$$\tau_r(B) = \left(\sum_{j > r} s_j(B)^2 \right)^{1/2}, \quad (4)$$

attained by the truncated singular-value decomposition.

Proof. Equation (1) makes $T(Z) = \|O^{1/2} Z\|_F$ a seminorm. The reverse triangle inequality and Theorem 2.1 give (3). Equation (4) is the Eckart–Young theorem [1]. $\square$

No normality, diagonalizability, phase arc, or one-step contraction is used. This is precisely why effective rank survives the RH-76 negative result.

3 Uniform effective-rank criterion

For a dyadic family, suppose RH-75 supplies M_k and $q_k < 1$. If there is a polylogarithmic rank schedule r_k and residuals $\tau_{r_k}(A_k^{M_k} X_k)$ such that

$$\sqrt{\frac{\|O_{k,M_k}\|}{1 - q_k^2}} \tau_{r_k}(A_k^{M_k} X_k) \leq C(1 + k)^a, \quad (5)$$

then the complete future differs from an r_k -dimensional source by at most a polylogarithmic amount. If the reduced source itself has a polylogarithmic Hardy bound, Stage A1 follows without global phase compression.

The attractive numerical possibility is stronger: a fixed $r = 4$ may suffice. RH-68 forbids inferring fixed Krylov depth from stability alone, but it does not forbid a physical family from producing fixed postblock singular rank.

4 Exact-dyadic interval audit

The frozen production arrays are the exact dyadic inputs certified in RH-70 [2]. For each channel, the audit:

1. computes $A^M X$ in 160-bit Arb arithmetic;
2. forms binary64 SVD candidates of ranks 1, 2, 4;
3. embeds each candidate as an exact dyadic matrix and recomputes its residual against the Arb state;
4. accumulates $\sum_{r < M} \|Y A^r\|_F^2 \geq \|O_M\|_2$ in Arb;
5. applies (3) with RH-70's certified block q .

Table 1: Validated minimum rank-two/rank-four energy capture over the two channels and maximum postblock participation rank.

σ	M	participation rank	rank-2 capture	rank-4 capture
0.16	4	1.004	0.999905	> 0.999999999999
0.08	9	1.218	0.999997	> 0.999999999999
0.04	16	1.438	0.999999998	> 0.999999999999
0.02	25	1.766	0.990845	> 0.999999999999
0.01	32	1.868	0.993404	0.9999999920

The rank-two loss is at most 0.916%, and rank four reduces the loss below 8.0×10^{-8} at the finest scale. The full-future transfer is equally small:

Table 2: Validated rank-four perturbation of the complete future Hardy energy.

σ	left	right
0.16	$5.05 \cdot 10^{-9}$	$4.40 \cdot 10^{-9}$
0.08	$3.40 \cdot 10^{-9}$	$3.01 \cdot 10^{-9}$
0.04	$2.76 \cdot 10^{-8}$	$3.48 \cdot 10^{-8}$
0.02	$5.47 \cdot 10^{-10}$	$6.77 \cdot 10^{-10}$
0.01	$1.86 \cdot 10^{-6}$	$5.35 \cdot 10^{-6}$

Even the worst value is negligible compared with the order-one Hardy energies and below one percent of RH-74’s smallest bridge slack.

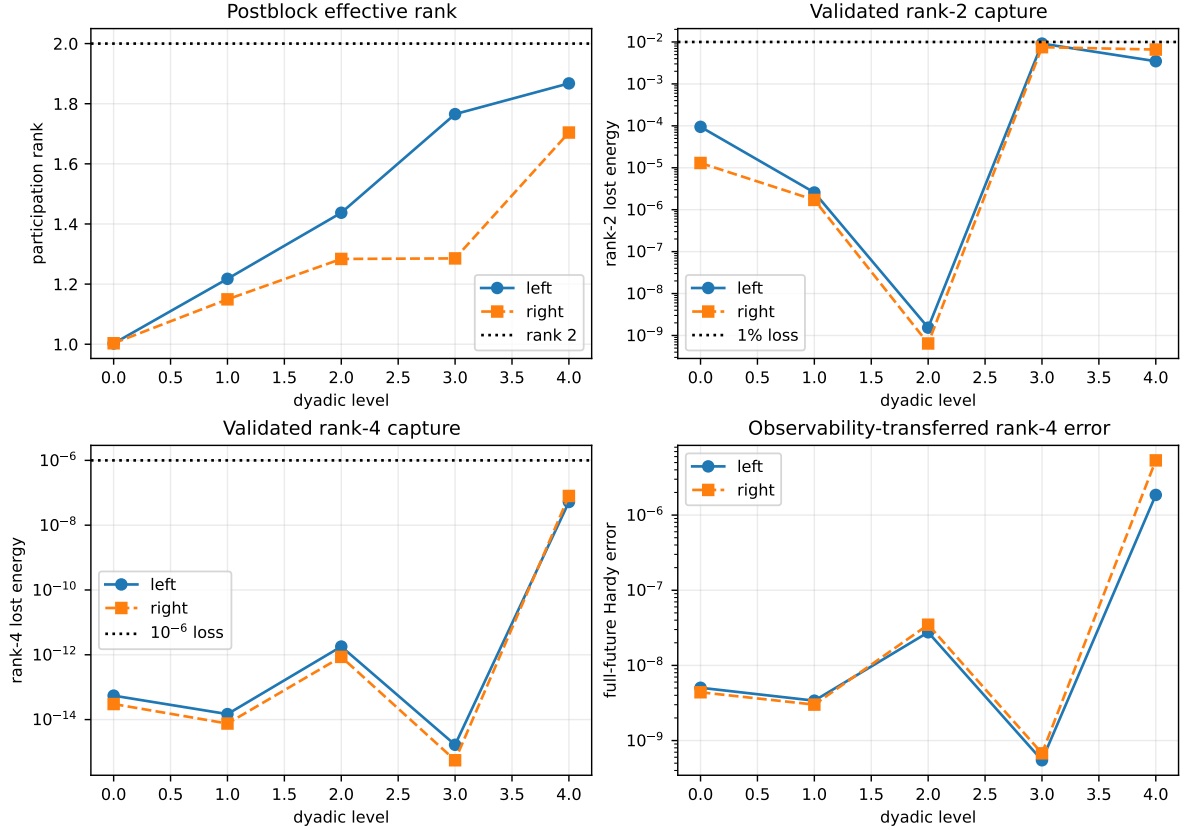


Figure 1: Postblock participation rank, validated rank-two/rank-four losses, and observability-transferred full-future error.

5 Route consequence

The maze now has a more plausible corridor. Raw phase support grows almost with dimension, but one physical block focuses the source columns into two to four singular directions. This is compatible with broad phases because the focusing includes radial decay and nonnormal column alignment.

The next theorem should no longer seek one shrinking arc. It should prove an all-level estimate of the form

$$s_5(A_k^{M_k} X_k) \leq \varepsilon_k \left\| A_k^{M_k} X_k \right\|_F,$$

with ε_k summable or polylogarithmically controlled, together with a bound on (5). RH-78 will formulate the conditional Stage A1 composition using RH-75’s block law and the present rank criterion.

This paper validates frozen finite matrices only. It does not prove uniform analytic effective rank, close Stage A1 or unconditional Stage A4, construct a renormalized determinant or Hilbert–Polya operator, derive a $T \log T$ law or prime-power trace formula, identify zeta zeros, or prove the Riemann Hypothesis.

6 Conclusion

Single-arc compression failed, but postblock singular compression succeeds decisively. A rank-four state reproduces the complete future Hardy response to within 5.35×10^{-6} at every archived scale. Effective rank is therefore the strongest current candidate for the uniform family mechanism.

References

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